

Bar Graphs to go with Within-Groups ANOVA

Analyze → Graphs → Legacy Dialogs → Bar

As for the WG ANOVA...

- Each quantitative variable (clark_age & lois_age) holds the DV for one of the IV conditions

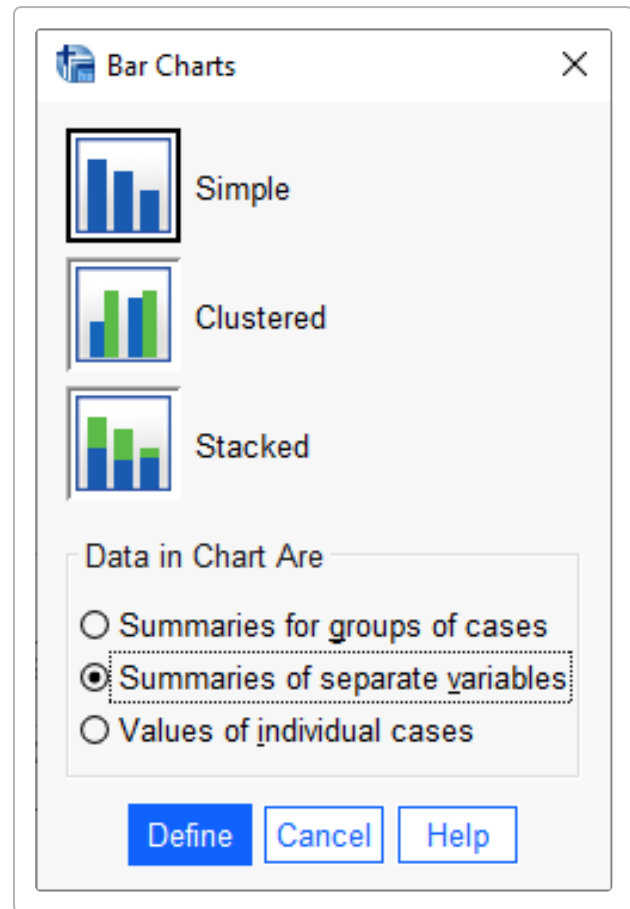
#1 Select type of chart

- Click on “Simple” icon
- Click the “Summaries of separate variables” radio button
- Click “Define”

For within-groups designs, use “Summaries of separate variables” — **not** “Summaries for groups of cases” (which is for between-groups designs).

#2 Select the Variables

- Highlight the variables
- Click the arrow to move the DV into the “Variable” window



#3 Write the Title

- Click on “Titles”
- Type titles, subtitles & footnotes as desired
- Click “Continue”

#4 Select the Whiskers

- Click Options
- Check “Display error bars”
- Check “Standard Error” or “Confidence intervals”
- Click Continue

Titles

Title

Line 1: Rotten Tomatoes Score from Critics versus Audiences

Line 2:

Subtitle:

Footnote

Line 1:

Line 2:

Continue Cancel Help

Options

Missing Values

☒ Exclude cases listwise

☐ Exclude cases variable by variable

☐ Display groups defined by missing values

☐ Display chart with case labels

☒ Display error bars

Error Bars Represent

☒ Confidence intervals

Level (%): 95.0

☐ Standard error

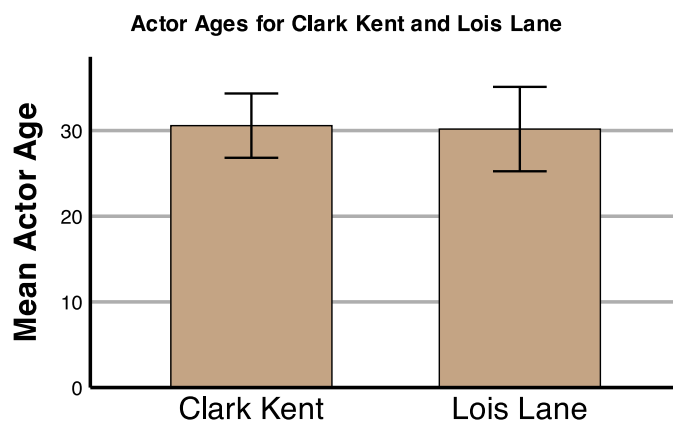
Multiplier: 2

☐ Standard deviation

Multiplier: 2.0

Continue Cancel Help

SPSS Output



Error bars: 95% CI

Interpreting the Graph

For within-groups designs, each bar represents the **mean** of one condition. In this example:

- The **Clark Kent bar** shows the mean age of actors who have played Clark Kent
- The **Lois Lane bar** shows the mean age of actors who have played Lois Lane

The **error bars** (vertical lines extending from each bar) show the **variability within each condition** — specifically, the 95% confidence interval around each mean. Wider error bars indicate more variability in the data, while narrower bars indicate consistency.

#5 Using the graph

Right-click the graph in the SPSS output window and select “Copy” to put the graph into a Word or other file.